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Research Paper

Enhancing the prediction of publications' long-term impact using early citations, readerships, and non-scientific factors

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ABSTRACT

This study aims to improve the accuracy of long-term citation impact prediction by integrating early citation counts, Mendeley readership, and various non-scientific factors, such as journal impact factor, authorship and reference list characteristics, funding and open-access status. Traditional citation-based models often fall short by relying solely on early citations, which may not capture broader indicators of a publication's potential influence. By incorporating non-scientific predictors, this model provides a more nuanced and comprehensive framework that outperforms existing models in predicting long-term impact. Using a dataset of Italian-authored publications from the Web of Science, regression models were developed to evaluate the impact of these predictors over time. Results indicate that early citations and Mendeley readership are significant predictors of long-term impact, with additional contributions from factors like authorship diversity and journal impact factor. The study finds that open-access status and funding have diminishing predictive power over time, suggesting their influence is primarily short-term. This model benefits various stakeholders, including funders and policymakers, by offering timely and more accurate assessments of emerging research. Future research could extend this model by incorporating broader altmetrics and expanding its application to other disciplines and regions. The study concludes that integrating non-citation-based factors with early citations captures a more complex view of scholarly impact, aligning better with real-world research influence.

1. Introduction

Accurately assessing the long-term impact of scientific publications is essential for understanding knowledge development, evaluating research contributions, guiding funding allocation, and shaping scientific policies and strategies. Traditionally, citation counts have been the primary measure of a publication's influence (Abramo, 2018; Bornmann & Daniel, 2008), with early citations often serving as a key indicator of future impact. As citations accumulate over time, they collectively represent the scholarly influence of a publication throughout its life cycle[1] (from date of publication to date of last citation), which can span decades.

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[1] The life cycle of a research product refers to the stages it undergoes in terms of citations, starting from its publication and continuing until its eventual decline, when it is no longer cited.

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However, waiting for decades to fully assess the impact of a publication is impractical for decision-makers who rely on timely evaluations to guide research investments and policy decisions. Although extending the citation-time window improves the accuracy of impact assessments, it comes at the cost of delayed insight. As a result, there is a natural tradeoff between timeliness and accuracy. This challenge calls for bibliometricians to improve the predictive power of early citation data, allowing for more immediate yet reliable assessments of long-term impact.

Scientific impact is a complex and dynamic process influenced by many factors beyond just early citations. Ideally, citations should reflect the scientific merit of a publication; however, empirical studies have demonstrated that various non-scientific factors also influence citation patterns (Kousha & Thelwall, 2024b; Mammola et al., 2022). By "non-scientific factors," we refer to any characteristic of a publication that is unrelated to its scientific content.

Recent advances in bibliometrics and data analytics have opened new avenues for refining models that forecast the long-term influence of publications. Various methods have been proposed to enhance early citation predictions, falling broadly into four categories: mathematical statistics, traditional machine learning, deep learning, and graph theory (Xia et al., 2023). The forecasting approach introduced in this study aligns with the first category.

In addition to early citation trends, non-scientific factors such as the journal's prestige, authors' institutional affiliations, collaboration networks, and media coverage have gained recognition as important drivers of a publication's visibility and impact over time (Mammola et al., 2022; Tahamtan, Safipour Afshar & Ahamdzadeh, 2016; Xie et al., 2019). We aim to develop a more accurate and robust predictive model by integrating these non-scientific factors with early citation data. This enhanced model will provide valuable tools for a wide range of stakeholders—including researchers, funders, and policymakers—enabling them to make more informed decisions about emerging research and to allocate resources more effectively.

We hypothesize that incorporating a diverse set of predictors will yield more accurate forecasts than models relying solely on early citation counts. Our approach seeks to deepen the understanding of the complex factors that drive research impact and to establish a comprehensive framework for evaluating how scientific work gains influence over time. This framework also offers scholars and practitioners a more precise tool for predicting the long-term scholarly impact of publications, even within short citation time windows. Ultimately, our goal is to provide scholars and practitioners with a set of weights for both early citations and non-scientific factors, whose combination would enhance the predictive power of early citation-based models. In particular, we try to answer the following research questions:

- What role do features beyond early citations play in predicting a publication's long-term impact?
- How does the influence of these predictors change with the length of the citation time window?
- How much does the predictive power of early citations improve when these additional predictors are also considered?

To address these questions, we will employ an OLS model, using the scholarly impact of Italian publications from 2010 to 2012 in the Web of Science (WoS) as the response variable, represented by citations received 11 years post-publication. The independent variables will include short-term impact, measured by citations within varying time windows of 0 to 6 years, readership metrics, and various non-scientific characteristics related to the publication's authorship, content, and venue.

In the following section, we review the relevant literature on late citation forecasting and the role of non-scientific factors. Section 3 outlines the field of observation, data sources, and methods employed. Results are presented in Section 4, followed by discussion, conclusions, and suggestions for future research in Section 5.

2. Review of the literature

Predicting the long-term impact of scientific publications is a very active area of research, driven by the increasing volume of scientific output and the need to evaluate its research influence. The breadth of the literature is so extensive that any attempt to synthesize it in a two-page review is bound to overlook some important contributions to the field. For a more comprehensive and in-depth analysis, readers are encouraged to consult: i) Hou et al. (2019), who reviewed various methods and applications for predicting publication impact; ii) Bai et al. (2017), who focused specifically on methodologies for predicting the impact of scholarly articles; and iii) Xia, Li and Li (2023) who offered a taxonomy of approaches. Additionally, they updated and expanded previous surveys by incorporating the latest advances in machine learning, including deep learning, recurrent neural networks (RNNs), convolutional neural networks (CNNs), and graph neural networks (GNNs).

Several methodologies and factors contribute to understanding and predicting the future impact of a publication, typically measured by citation counts or other indicators of scholarly recognition. Below is a review of key concepts, models, and approaches used in the literature to predict the long-term impact of scientific publications.

In this section, we organize the literature into two main thematic areas: i) Citation-based prediction models, and ii) Features influencing citation impact. We conclude by synthesizing gaps in the literature that motivate our study.

2.1. Citation-based prediction models

Early models of citation prediction primarily relied on citation dynamics and mathematical frameworks. The assumption that early citation counts predict long-term impact is well established (Bornmann & Daniel, 2010). Time series models such as exponential decay and autoregression (Wang et al., 2013; Yu et al., 2014) have been widely applied to simulate citation trajectories.

Other statistical approaches include survival analysis to assess the longevity of citation activity (Eom & Fortunato, 2011; Parolo

et al., 2015). Preferential attachment models, often referred to as the "rich-get-richer" principle, also offer insights into cumulative advantage in citation growth (Newman, 2018; Price, 1976). Cross-disciplinary studies (Garfield, 2006; Wang et al., 2021) highlight substantial variance in citation patterns across fields, necessitating domain-aware modeling strategies (Abramo et al., 2019).

The emergence of machine learning has enriched prediction capabilities. Techniques range from traditional models (random forests, SVMs) to deep learning (CNNs, RNNs, DNNs) (Acuna et al., 2012; Himani et al., 2022; Li et al., 2019). NLP tools enable content analysis by extracting features such as abstract complexity or topical novelty (Yau et al., 2014).

Integrated models that combine citation data with altmetrics and author-level information have shown superior performance (Qiu & Han, 2024; Zhang & Wu, 2024). These hybrid models can be adapted to disciplinary idiosyncrasies, enhancing generalizability.

Comparative assessments of model performance show that more complex models (e.g., neural networks, domain-specific hybrids) tend to outperform basic regression or citation-only approaches (Mazloumian et al., 2011; Penner et al., 2013; Kosteas, 2020).

2.2. Features influencing citation impact

Numerous structural, content-based, and contextual features affect a publication's citation trajectory. Document type plays a key role—review articles generally receive more citations than original research or proceedings (Aksnes, 2006; Moed, 2005). Content novelty and interdisciplinarity have also been linked to higher long-term impact (Porter & Rafols, 2009; Uzzi et al., 2013). Disruptiveness, measured by deviation from existing paradigms, emerges as a meaningful citation predictor (Foster et al., 2015; Leahey et al., 2023). Papers that span multiple fields are often cited across different domains, leading to a more sustained and broader citation impact (Abramo et al., 2017).

Author-related features such as collaboration network centrality, institutional affiliation, and prior productivity influence impact (Sarigöl et al., 2014; Petersen et al., 2014; Mazloumian et al., 2011). The number of authors is positively associated with citation counts (Abramo & D'Angelo, 2015; Fox, Paine & Sauterey, 2016; Talaat & Gamel, 2023), as are the number of affiliations and countries represented (Glänzel & De Lange, 2002; Narin & Whitlow, 1990; Sanfilippo, Hewitt & Mackey, 2018). Publications authored by native English speakers also tend to receive more citations (Hyland, 2016; Ammon, 2012; Meneghini & Packer, 2007; Van Leeuwen & Moed, 2002).

Longer reference lists have been found to correlate with increased citation counts, especially when they include highly cited and recent publications (Alimohammadi & Sajjadi, 2009; Fox, Paine & Sauterey, 2016; Jiang, He & Ni, 2013; Liu et al., 2022; Mammola et al., 2021; Sivadas & Johnson, 2015). Additional factors include the diversity of cited fields and the cognitive distance of these citations (Wang, Thijs, & Glänzel, 2015), as well as the proportion of self-citations (Ruan et al., 2020).

The prestige of the publication venue, often proxied by journal impact factor, is another strong correlate (Traag, 2021; Mammola et al., 2021; Levitt & Thelwall, 2011), though its reliability has been debated (Larivière & Gingras, 2010).

Additional predictors include accessibility, as open access articles generally attract more citations (Gargouri et al., 2010; Wang et al., 2015; Yu et al., 2022), and paper length or number of references, which have been associated with greater visibility and perceived comprehensiveness (Liu et al., 2022; Fox et al., 2016; Ball, 2008).

Linguistic characteristics such as clarity of title and abstract also affect discoverability and impact (Abramo et al., 2016; Heßler & Ziegler, 2022; Rossi & Brand, 2020). Papers with international co-authorship or authors affiliated with English-speaking institutions tend to receive more citations (Hyland, 2016; Wuchty et al., 2007). Funding, while generally supportive of citation growth (Rigby, 2013; Wang et al., 2017), may have a diminishing effect over time. Early visibility through social media (Özkent, 2022) and Mendeley readership has proven useful for anticipating long-term influence (Haustein et al., 2014; Mohammadi & Thelwall, 2014; Zahedi, Costas & Wouters, 2014). Finally, studies indicate that financial support often correlates with higher citation rates (Aksnes, 2003; Boyack & Jordan, 2011; Lewison & Dawson, 1998; Rigby, 2013; Wang, Veugelers & Stephan, 2017)

Although many of these predictors show consistent effects, they are often considered in isolation and underutilized in conventional models due to limitations in data accessibility and integration. Moreover, interactions among variables and the temporal dynamics of their influence are frequently overlooked.

Despite the progress in prediction models and identification of impact-driving features, existing research often treats predictors in isolation. Most studies emphasize a narrow subset of variables (typically early citations and journal impact) without integrating broader contextual features. Moreover, the relative predictive power of different features at varying citation time windows is underexplored.

This study addresses these gaps by integrating a broad array of non-scientific features with early citation and readership data using a structured, statistical modeling approach. We aim to provide a more nuanced and temporally robust framework for forecasting the impact of long-term publications.

In the following analysis, we included all non-scientific factors that (i) empirical studies have shown to be associated with citation outcomes, and (ii) can be readily and reliably measured.

3. Data and methods

Our analysis focuses on 2010–2012 Web of Science (WoS) publications in which at least one author lists "Italy" as their affiliation country, with document types limited to articles, reviews, and conference proceedings. These publications were sourced from the National Citation Report-Italy through a licensing agreement with Clarivate Analytics. Access to this report enables us to obtain predictors such as citation counts at various intervals post-publication, financial support information, access type, number of references, etc. The dataset includes 293,538 publications in all. Through the WoS classification schema, each of them is assigned to the

subject category (SC) of the hosting journal.[2] For each publication, we extract information regarding various features, including citation impact, characteristics of the byline, attributes related to the content and publication venue, and features associated with the publication's bibliography.

We use an OLS model to examine the role of features beyond early citations in predicting a publication's long-term impact. The normalized impact (IMPACT_*ti*) is measured by normalized citations, defined as the citation count divided by the average citations of all Web of Science (WoS) publications within the same subject category (SC) and publication year. Citations assessed 11 years post-publication serve as our response variable for long-term impact. Short-term impact is calculated over varying citation windows from 0 to 6 years. For each time window, we also compute normalized Mendeley readership counts (READ_*ti*), scaled to the average readership of WoS publications in the same SC and publication year.

Additional features related to the publication's content and venue include:

- Authorship: Number of authors (AUTH) and a dummy variable for native English-speaking authors (ENG) to account for potential linguistic advantage.
- International collaboration: A dummy variable for publications with multiple countries listed in the affiliations (FOREIGN).
- Funding: A dummy variable for funded research, based on acknowledgments (FUNDING).
- Open access status: A dummy variable for Green, Hybrid, and Gold open access publications (OPEN).
- Length: Number of pages (PAGES).
- References: Reference list length as the count of cited references (REFER).
- Journal impact: Normalized impact factor of the source journal in the publication year (IF).
- Disciplinary classification: 247 dummy variables (D_SUBCATj) representing the WoS SCs in the sample.
- Document type: Two dummy variables distinguishing articles (ART) and reviews (REWs) from the baseline, conference proceedings.

The list of variables included in the dataset is summarized in Table 1.

As discussed in the upcoming section on descriptive statistics, several variables exhibit high variability relative to the mean, strongly influenced by the presence of outliers. A preliminary analysis using local polynomial smoothing indicates that, for most variables, the relationships appear linear on a logarithmic scale. Consequently, we applied a base-10 logarithmic transformation to integer and continuous variables, adding a unit to prevent data loss due to zero values. This approach aligns with prior literature recommendations for variables such as IMPACT_*ti* (Bornmann & Leydesdorff, 2017), READ (Costas, Perianes-Rodríguez & Ruiz-Castillo, 2017), AUTH (De Santis Puzzonia, Beltram, Cicero & Malgarini, 2018), PAGES, IF, and REFER (Nicolaisen & Frandsen, 2021).

To estimate long-term impact—specifically, 11 years post-publication—as a function of available features, we implemented a series of regression models, summarized in Table 2. This table includes 21 regression models divided into three groups: "Full," "Reduced," and "Completely Reduced" models, each comprising seven iterations.

In the Full models, the dependent variable is the 11-year impact (IMPACT_t11), with a set of independent variables that combines time-varying and time-invariant factors. The time-varying vector X, which includes IMPACT and READ, is adjusted in each iteration. For example, in the first Full model, we use IMPACT_t0 and READ_t0; in the second, IMPACT_t1 and READ_t1; continuing up to IMPACT_t6 and READ_t6 in the seventh model. Other regressors—Z (non-binary variables such as AUTH, REFER, IF, PAGES), D (binary variables such as D_ART, D_ENG, D_FOREIGN, D_REW, D_FUNDING, D_OPEN), and S (subject categories)—remain constant across iterations.

The same dependent variable (IMPACT_t11) is used in the Reduced models but with fewer regressors. Here, only the variable IMPACT is updated across iterations, with IMPACT_t0 used in the first model, IMPACT_t1 in the second, and so on up to IMPACT_t6. The only time-invariant variable, IF, along with control variables S (subject categories), remains constant throughout. These models examine the effects of early citations and journal impact factor alone, facilitating comparison with similar analyses in the literature (Abramo, D'Angelo & Felici, 2019) and comparing it with the Full model highlights differences in predictive power.

The final set of models, the Completely Reduced models, streamlines the regressor set further. Here, only the impact variable (IMPACT_t) and control variables S (subject categories) are included, with the aim of identifying their effects. This simplified approach is widely used due to the accessibility of early citation data, and comparing it with the Full model highlights differences in predictive power.

4. Results

4.1. Descriptive statistics and correlation analysis

Table 3 presents the descriptive statistics of the variables used in this study, providing an overview of their distribution and key characteristics. The mean values of dummy variables represent the percentage of publications characterized by the specific attribute. Accordingly, 21.9 % of publications have at least one native English-speaking author; in 44 % of cases, the address list contains more

[2] Publications hosted in multicategory journals are fully assigned to all SCs of the journal.

Table 1
– Variable description and related acronym.

Variables Description	Variable type	Acronym
Publications' impact after *i* years	Continuous	IMPACT_t*i*
Mendeley readerships after *i* years	Continuous	READ_t*i*
Number of authors in the byline	Integer	AUTH
Presence of at least one native English-speaking author in the byline	Dummy	D_ENG
Two or more countries in the affiliation list	Dummy	D_FOREIGN
Publication receiving financial support/funding aknowledgement	Dummy	D_FUNDING
Open access publication	Dummy	D_OPEN
Publication length (number of pages)	Integer	PAGES
Normalized impact factor of the hosting source	Continuous	IF
Document type - Article	Dummy	D_ART
Document type – Review	Dummy	D_REW
Number of references of the publication	Integer	REFER
SC of the publication	247 dummies	D_SUBCAT_j

Table 2
– Models estimated and their iterations.

Model	Equations	Iterations
Full	$Y_{i(11)} = \beta_1 X_{i,t} + \beta_2 Z_i + \beta_3 D_i + \beta_4 S_i + e_{i(11)}$ Where: Y = impact value of publication i at time 11 (IMPACT_t11); β_1 = vector of regression coefficients for the vector of variables that are replaced in each model (specifically composed of the variables IMPACT_t and READ_t); t is the time, with $0 \leq t \leq 6$; β_2 = vector of regression coefficients for the vector of variables Z that do not vary over time (e.g. AUTH, REFER, IF, PAGES); β_3 = vector of regression coefficients for the vector of binary variables D that do not vary over time (e.g., D_ART, D_REW, D_ENG, D_FOREIGN, D_FUNDING, D_OPEN); β_4 = vector of regression coefficients for the vector of control variables S, representing the SC in which each publication i is classified; $e_{i(11)}$ = error term for publication i at time $t = 11$.	7 iterations (t_0–t_6): Regressors characterized by Z, D, and S remain constant, but the two variables X are replaced in each iteration (i.e. the vector X is characterized by IMPACT and READ. In the first model, we include IMPACT_t0 and READ_t0; in the second model, IMPACT_t1 and READ_t1 are introduced, and so on, up to IMPACT_t6 and READ_t6 in the seventh model.)
Reduced	$Y_{i(11)} = \beta_1 X_{i,t} + \beta_2 IF_i + \beta_3 S_i + e_{i(11)}$ Where: Y = impact value of publication i at time 11 (IMPACT_t11); β_1 = regression coefficient for the variable that is replaced in each model (specifically IMPACT_t) t is the time, with $0 \leq t \leq 6$; $\beta_2 IF_i$ = regression coefficient for the variable IF that does not vary over time β_3 = vector of regression coefficients for the vector of control variables S, representing the SC in which each publication i is classified; $e_{i(11)}$ = error term for publication i at time $t = 11$.	7 iterations (t_0–t_6): Regressors characterized by IF and S remain constant, but the one variable X is replaced in each iteration (i.e. X is characterized only by IMPACT. In the first model, we include IMPACT_t0; in the second model, IMPACT_t1 and so on, up to IMPACT_t6 in the seventh model.)
Completely Reduced	$Y_{i(11)} = \beta_1 X_{i,t} + \beta_2 S_i + e_{i(11)}$ Where: Y = impact value of publication I at time 11 (IMPACT_t11); β_1 = regression coefficient for the variable that is replaced in each model (specifically IMPACT_t0) t is the time, with $0 \leq t \leq 6$; β_2 = vector of regression coefficients for the vector of control variables S, representing the SC in which each publication i is classified; $e_{i(11)}$ = error term for publication i at time $t = 11$.	7 iterations (t_0–t_6): Regressors characterized by S remain constant, but the one variable X is replaced in each iteration (i.e. X is characterized only by IMPACT. In the first model, we include IMPACT_t0; in the second model, IMPACT_t1 and so on, up to IMPACT_t6 in the seventh model.)

than one country; 33.7 % of publications are Open Access; and 49 % have received financial funding. Most of the publications are classified as articles (83.8 %), while the remaining publications are reviews (6.1 %) and conference proceedings (10.1 %), which are the residual category not included in the dummy variables.

The 293,538 publications are classified into 248 distinct SCs.[3] These categories are highly heterogeneous, ranging from those with as few as two publications to a maximum of 7448, with an average of 1183.6 observations. The Herfindahl-Hirschman Index, which measures the concentration of publications within a single category, is 0.011. This indicates a highly fragmented corpus of

[3] Compared to the 254 total SCs in the WoS schema, during the observed period (2010-2012), there are no publications by Italian authors in 6 Art & Humanities SCs (Dance; Folklore; Literature, African, Australian, Canadian; Literature, German, Dutch, Scandinavian; Literature, Slavic; Poetry).

Table 3
- Descriptive statistics of analyzed variables (293,538 observations).

Variables	Mean	Std Dev.	Min	Max
IMPACT_t11	1.057	2.553	0.000	309.625
IMPACT_t6	0.936	1.980	0.000	196.955
IMPACT_t5	0.928	1.902	0.000	193.313
IMPACT_t4	0.917	1.837	0.000	210.344
IMPACT_t3	0.897	1.773	0.000	227.133
IMPACT_t2	0.855	1.704	0.000	239.548
IMPACT_t1	0.730	1.477	0.000	193.558
IMPACT_t0	0.323	0.879	0.000	78.314
READ_t6	1.664	4.220	0.000	951.598
READ_t5	1.699	4.662	0.000	1137.025
READ_t4	1.722	4.957	0.000	1190.991
READ_t3	1.701	4.859	0.000	1077.796
READ_t2	1.592	4.485	0.000	967.132
READ_t1	1.307	3.071	0.000	487.578
READ_t0	0.719	1.602	0.000	216.927
AUTH	13.849	127.614	1.000	3221.000
D_ENG	0.219	0.413	0.000	1.000
D_FOREIGN	0.440	0.496	0.000	1.000
D_FUNDING	0.490	0.500	0.000	1.000
D_OPEN	0.337	0.473	0.000	1.000
PAGES	9.866	8.176	1.000	1504.000
IF	1.262	1.139	0.000	21.070
D_ART	0.838	0.368	0.000	1.000
D_REW	0.061	0.239	0.000	1.000
REFER	38.465	36.365	0.000	5340.000

publications, with no single SC holding a significant share of the total publications. These variables will be used as control variables in the subsequent analysis, to account for the variability of citation patterns across SCs.

As previously indicated, the available impact values correspond to distinct and specific time points, allowing us to differentiate between short-term impact (within 6 years of publication) and long-term impact (11 years after publication). As might be expected, these impact values are highly correlated, as demonstrated in Fig. 1 (left panel). Therefore, we exclude from the outset any predictive model that considers all impact variables simultaneously, as such a model would be vulnerable to severe multicollinearity issues, which would compromise the accuracy of the predictive estimates.

As observed, the correlation between impact values at successive years is very high, with an increasing trend as the years progress. Specifically, the correlation between the impact in the publication year (IMPACT_t0) and the impact in the following year (IMPACT_t1) is 0.77. This value rises to 0.96 between t1 and t2, 0.98 between t2 and t3, and stabilizes at 0.99 for subsequent time points up to 6 years.

In contrast, the long-term impact, measured 11 years after publication, shows a much weaker correlation with the impact in the publication year (0.39). However, the correlation between the long-term impact and the impact two years after publication increases to 0.72.

Similar considerations can be made when analyzing the correlation of readership measured at different time points (Fig. 1– right panel). There is consistently a strong correlation between consecutive time points, and the correlation decreases as t increases: readership at $t = 0$ (READ_t0) is correlated with readership at $t = 1$ with a value of 0.82. As we fix READ_t0, the correlation diminishes progressively over time, reaching 0.67 six years after publication.

In general, the correlation values are higher, confirming that Mendeley readership exhibits both rapid growth and long-term sustainability from the early stages of the publication lifecycle (Fang, Ho, Han & Wu, 2024).

In addition to the autocorrelations of impact and readership at different time points, we analyzed all correlations between the time-invariant variables related to publication features (Fig. 2).

No correlations were sufficiently high to suggest potential multicollinearity in constructing the statistical model, but all are statistically significant. The highest correlation was observed between the dummy variable indicating when the address list contains more than one country (D_FOREIGN) and the dummy variable indicating the presence of a native English-speaking author (D_ENG), with a value of 0.60.

The variable related to the document type 'Review' (REW) is correlated with the number of references (0.39) and inversely correlated with the dummy variable defining publications as articles (ART), as these two categories are mutually exclusive. Lastly, the correlation between the number of references (REFER) and the number of pages (PAGES) is 0.36. No other notable correlations were identified.

4.2. OLS models

Table 3 presents the coefficients of a linear regression model across seven iterations (t_0-t_6) for the "Full" model, offering insights into the evolving significance of various explanatory variables over time. The results show that the coefficient for L_IMPACT_t steadily

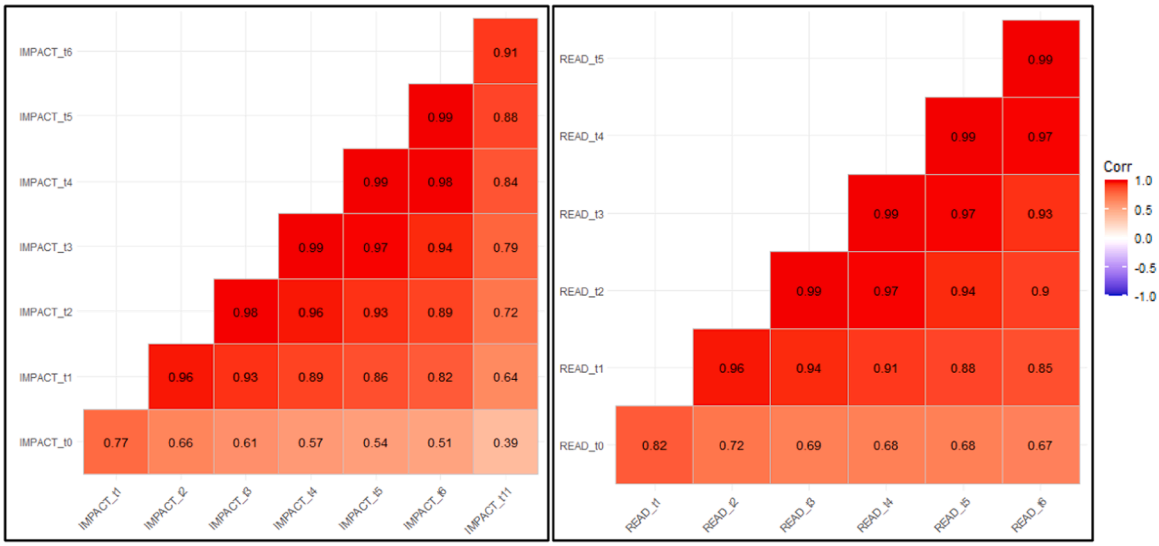

Fig. 1. Autocorrelation matrix of impact variables (left panel) and readership (right panel), at time t.

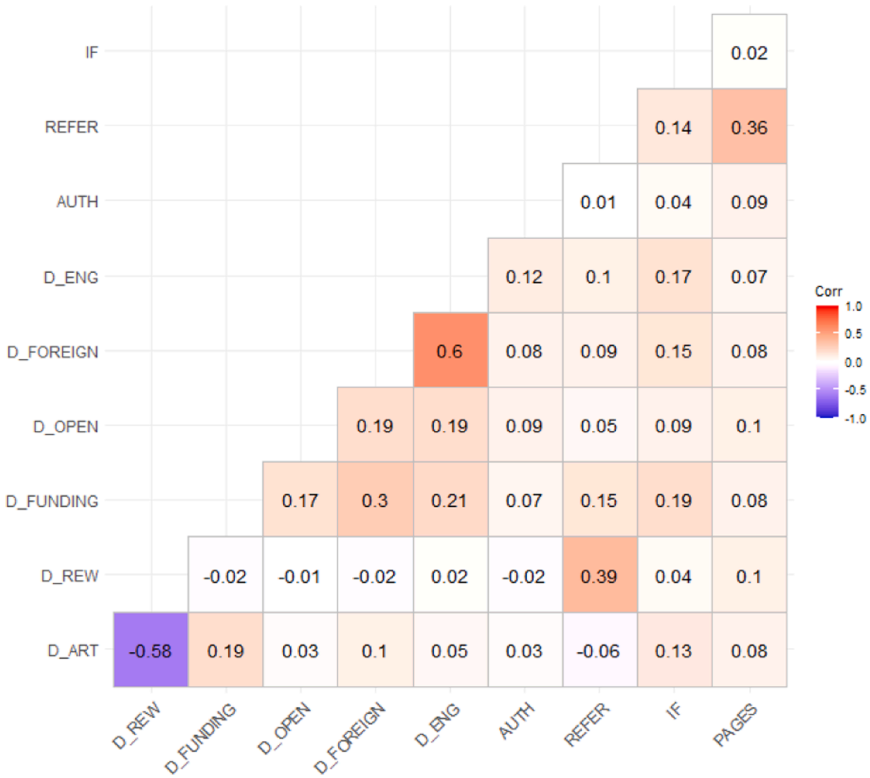

Fig. 2. Matrix of correlations among variables.

increases from 0.243 at t_0 to 0.976 at t_6, indicating that the impact of this variable becomes progressively more significant with each iteration. The constant statistical significance (denoted by ***) across all time points underscores the robustness of its influence throughout the model's development.

Conversely, L_READ_t shows a decreasing trend, starting at 0.278 at t_0 and dropping to 0.070 at t_6. This suggests a diminishing role of readings over time, although the variable remains statistically significant in every iteration, indicating that it continues to contribute to the model, albeit with less influence as the years go by. Similarly, L_AUTH begins with a positive coefficient at t_0 (0.046) but turns negative from t_2 onwards, stabilizing at approximately −0.017. This shift suggests that the impact of the number of authors becomes

Table 3a
– Regression coefficients and diagnostic statistics about the Full models, with seven iterations (from $t = 0$ to $t = 6$).

Variables	$t = 0$	$t = 1$	$t = 2$	$t = 3$	$t = 4$	$t = 5$	$t = 6$
L_IMPACT_t	0.243***	0.523***	0.717***	0.833***	0.899***	0.944***	0.976***
L_READ_t	0.278***	0.242***	0.187***	0.143***	0.115***	0.091***	0.070***
L_AUTH	0.046***	0.004***	−0.012***	−0.017***	−0.018***	−0.018***	−0.017***
D_ENG	0.035***	0.011***	0.002*	−0.001	−0.001	−0.002**	−0.002***
D_FOREIGN	0.019***	0.009***	0.005***	0.003***	0.002***	0.001*	0.001**
D_FUNDING	0.010***	−0.006***	−0.009***	−0.008***	−0.006***	−0.004***	−0.001***
D_OPEN	0.035***	0.017***	0.010***	0.008***	0.006***	0.005***	0.004***
L_PAGES	0.061***	0.050***	0.043***	0.034***	0.026***	0.020***	0.016***
L_IF	0.285***	0.122***	0.049***	0.017***	0	0.001***	−0.008***
D_ART	0.054***	0.041***	0.023***	0.015***	0.012***	0.010***	0.010***
D_REW	0.141***	0.059***	0.013***	−0.005**	−0.010***	−0.012***	−0.011***
L_REFER	0.095***	0.039***	0.011***	−0.001	−0.005***	−0.006***	−0.006***
Adjusted R^2	0.758	0.846	0.900	0.931	0.951	0.965	0.975
F-statistics	3542.771	6214.632	10,219.890	15,387.436	22,116.089	31,240.159	43,539.748
p-value F	0.000	0.000	0.000	0.000	0.000	0.000	0.000
RSE	0.363	0.289	0.233	0.193	0.163	0.138	0.117
DoF	293,279	293,279	293,279	293,279	293,279	293,279	293,279

* Control variables 247 SC dummies. SC number 1 is considered as the baseline– see supplemental material (SI-Appendix A.xlsx).

detrimental in later stages, with its influence weakening and eventually becoming counterproductive.

Something similar concerns the variable D_ENG, which starts with a small positive coefficient at t_0 (0.035) but turns negative by t_3 (−0.001) and further decreases to −0.002 at t_6. Initially, the "linguistic advantage" of having English-speaking co-authors in the byline appears to be in place, but this reverses over time. For D_FOREIGN, the coefficient declines from 0.019 at t_0 to 0.001 at t_6, indicating a reduced influence of the presence of almost another country as iterations progress, though it remains statistically significant.

In contrast, D_FUNDING demonstrates a negative trajectory, with the coefficient turning from 0.010 at t_0 to −0.006 at t_1 and further declining to −0.001 at t_6. This pattern suggests that funding exerts an increasingly negative influence, particularly in the later stages of the model.

D_OPEN shows a gradual decline from 0.035 at t_0 to 0.004 at t_6, retaining its statistical significance throughout, suggesting that while openness remains a positive contributor, its impact lessens over time. Similarly, L_PAGES exhibits a decreasing coefficient, moving from 0.061 at t_0 to 0.016 at t_6. Although its influence diminishes, it remains statistically significant, indicating that the number of pages continues to play a role, albeit a smaller one in later iterations.

The behavior of L_IF is noteworthy, starting with a significant positive coefficient at t_0 (0.285) but decreasing and eventually turning negative at t_6 (−0.008). This suggests that the impact factor (IF) initially contributes positively to the model but loses relevance and becomes a negative factor as iterations progress. The same trend is observed in D_ART, where the coefficient decreases from 0.054 at t_0 to 0.010 at t_6, reflecting a declining influence of being an article over time, though it remains statistically significant.

Both D_REW and L_REFER exhibit similar behavior. D_REW starts with a strong positive coefficient (0.141 at t_0) but turns negative at t_3 (−0.005) and continues to decrease to −0.011 at t_6. L_REFER follows a comparable path, starting at 0.095 at t_0 and turning negative at t_3 (−0.001), further declining to −0.006 at t_6. These results suggest that the initial positive impact of being a review and the number of references diminishes over time, and by the later iterations, these variables contribute negatively to the model.

The performance of the "Full model" improves significantly across iterations, as demonstrated by the adjusted R^2, which increases from 0.758 at t_0 to 0.975 at t_6. This indicates a substantial enhancement in the model's explanatory power as more iterations are considered. Similarly, the F-statistic increases markedly, from 3542.771 at t_0 to 43,539.748 at t_6, signaling a strong improvement in the model's overall significance. Additionally, the residual standard error (RSE) decreases progressively, from 0.363 at t_0 to 0.117 at t_6, reflecting a reduction in prediction error and an overall improvement in the accuracy of the model. The p-value for the F-statistic remains consistently at 0.000 across all iterations, confirming the statistical significance of the model at each stage.

The second table (Table 4) presents results from the Reduced model over seven iterations (t_0 - t_6), which includes fewer variables compared to the previous model.

In the Reduced model, L_IMPACT_t0 begins at a much higher value than the previous "Full model", 0.529 at t_0, and grows to 1.048

Table 4
– Regression coefficients and diagnostic statistics about the Reduced models, with seven iterations (from $t = 0$ to $t = 6$).

Variables	$t = 0$	$t = 1$	$t = 2$	$t = 3$	$t = 4$	$t = 5$	$t = 6$
L_IMPACT_t0	0.529***	0.790***	0.929***	0.993***	1.024***	1.040***	1.048***
L_IF	0.454***	0.200***	0.086***	0.038***	0.014***	0.003***	−0.001*
Adjusted R^2	0.711	0.828	0.893	0.928	0.949	0.964	0.974
F-statistics	2904.625	5683.263	9875.971	15,240.533	22,143.605	31,515.997	44,179.767
p-value F	0.000	0.000	0.000	0.000	0.000	0.000	0.000
RSE	0.396	0.305	0.240	0.197	0.166	0.140	0.119
DoF	293,289	293,289	293,289	293,289	293,289	293,289	293,289

* Control variables 247 SC dummies. SC number 1 is considered as baseline - see supplemental material (SI-Appendix B.xlsx).

at t_6. The impact of this variable is stronger from the beginning, and it also grows more quickly, achieving near-maximal values earlier in the iteration process. By t_4, the coefficient is already at 1.024, compared to 0.899 in the full model. In other words, L_IMPACT_t0 has a higher and more immediate impact on the outcome, indicating that in the absence of other variables, this factor exerts a more dominant influence earlier in the process.

L_IF follows a similar pattern in both models, but its influence diminishes more rapidly in the Reduced model. Without other variables, the impact factor contributes strongly to the early iterations but quickly loses importance.

Lastly, Table 5 presents results for the third model, where the regressors are limited to IMPACT only (across different time points from t_0 to t_6) and SC controls. This "Completely Reduced" model aims to isolate the effects of early citations and disciplinary differences by excluding the control for the impact factor of the hosting source. The coefficient for L_IMPACT at time 0 is very high (0.660) and the differences with the previous models tend to diminish as the time increases. At time 6, no differences are observed in the IMPACT coefficients between this model and the previous Reduced model.

4.3. Comparing OLS models

Fig. 3 presents a comparative analysis of the three regression models evaluated using Adjusted R-squared (Adjusted R^2) across various values of IMPACT_t. Adjusted R^2 serves as an indicator of the explanatory power of each model, accounting for the number of predictors included, and thus provides a more robust measure than the standard R^2, especially in models with multiple variables.

The comparison shows that the Full model consistently exhibits the highest Adjusted R^2 values across the first three iterations, confirming that it has the strongest explanatory capacity among the three models.

To compare the "Full" model with the "Reduced" model and, subsequently, with the "Completely Reduced" model, fourteen ANOVA tests were conducted, one for each variant of each model comparison. In this context, the ANOVA test evaluates whether the additional predictors included in the more complex models significantly improve their explanatory power. The F-statistic obtained from the ANOVA provides a measure of how much better the Full model explains the variability in the data compared to simpler models, helping to determine if the increase in model complexity is statistically justified. This statistic is given by the following ratio:

$$F = \frac{\left(\frac{RSS_{reduced\ (or\ completly\ reduced)} - RSS_{full}}{p}\right)}{\left(\frac{RSS_{full}}{n-p-1}\right)} \tag{1}$$

Where:

- $RSS_{Reduced\ (or\ Completely\ Reduced)}$ is the residual sum of squares of the "Reduced" (or "Completely Reduced") model,
- RSS_{full} is the residual sum of squares of the "full" model,
- p is the number of additional variables in the full model,
- n is the total number of observations.

A high F value indicates that the more complex model explains a greater portion of the variability in the data compared to the Reduced model. All comparisons yielded an F value with a p-value < 0.01, meaning that the "Full" model is consistently more significant overall than the "Reduced" and the "Completely Reduced" one. The F-statistic values are represented in Fig. 4 for each model variant.

We can observe that the greater predictive power of the full model decreases with the citation time window, as the contribution of non-scientific factor decreases in favor of that of early citations. In fact, the F-statistic starts at over 5000 in the ANOVA model comparing the Full and Reduced models and at over 10,000 in the ANOVA model comparing the Full and Completely Reduced models but drops significantly to below 1000 for both comparisons by t_6.

4.4. Post-estimation analysis

4.4.1. Normality of residuals

To assess the adequacy of the OLS regression models, it is crucial to examine the distribution of the residuals and ensure that the

Table 5
– Regression coefficients and diagnostic statistics about the Completely Reduced models, with seven iterations (from $t = 0$ to $t = 6$).

Variables	$t = 0$	$t = 1$	$t = 2$	$t = 3$	$t = 4$	$t = 5$	$t = 6$
L_IMPACT_t0	0.660***	0.861***	0.962***	1.007***	1.029***	1.042***	1.048***
Adjusted R^2	0.665	0.820	0.892	0.928	0.949	0.964	0.974
F-statistics	2345.407	5398.664	9769.051	15,239.953	22,215.850	31,641.589	44,357.504
p-value F	0.000	0.000	0.000	0.000	0.000	0.000	0.000
RSE	0.427	0.312	0.242	0.198	0.166	0.140	0.119
DoF	293,290	293,290	293,290	293,290	293,290	293,290	293,290

*Control variables 247 SC dummies. SC number 1 is considered as baseline - see supplemental material (SI-Appendix C.xlsx).

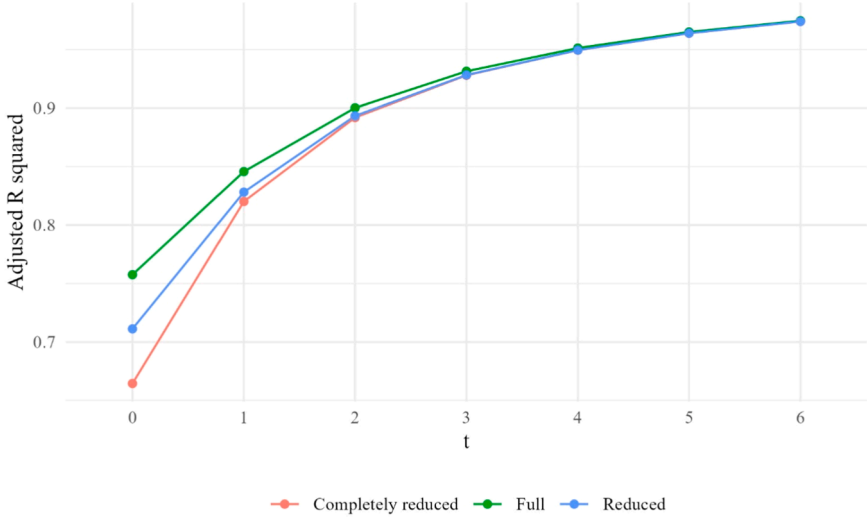

Fig. 3. Comparison of Adjusted R^2 statistics after OLS regressions for Full, Reduced models, and Completely Reduced models (for each citation time window t).

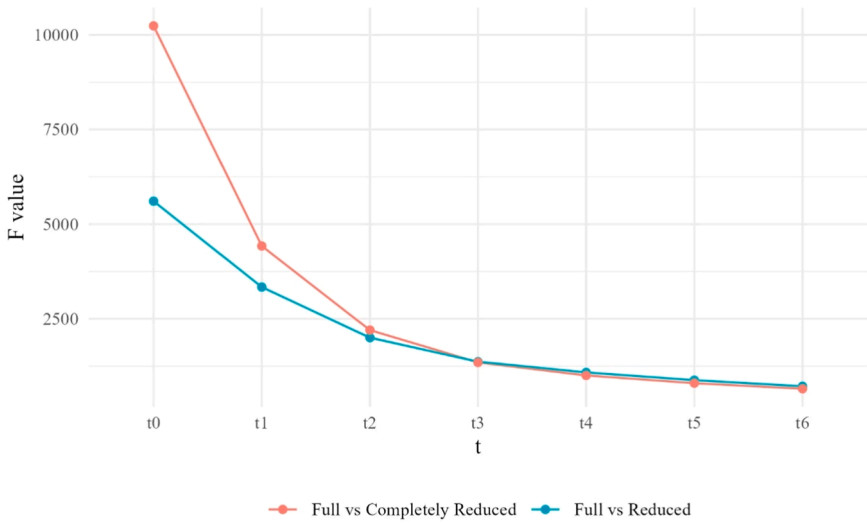

Fig. 4. Comparison of F values statistics after ANOVA among Full and Reduced models and among Full and Completely Reduced models (for each citation time window t).

assumptions of normality are met. Deviations from normality can affect the validity of hypothesis tests, the precision of confidence intervals, and the overall reliability of the model. Diagnostic plots such as histograms of residuals and Q-Q (Quantile-Quantile) plots are commonly used to evaluate this assumption. These plots provide visual insights into how well the residuals conform to the expected normal distribution.

For example, Fig. 5 illustrates the normality check for residuals for the Full model version at t_0. The histogram in the left panel displays the distribution of the residuals. It is overlaid with a kernel density estimation (red line), which provides a smoothed estimate of the data distribution. The goal of this plot is to compare the empirical distribution of the residuals to the expected normal distribution. The residuals follow a roughly normal distribution, with most values concentrated around zero. The bell-shaped curve of the kernel density (red line) fits the histogram reasonably well. While the residuals generally exhibit a regular shape, there are slight deviations at the tails, where the distribution does not perfectly align with the normal density. These deviations, though small, could indicate minor departures from normality in the residuals. However, some caution is advised due to the visible deviations, which might impact the model's assumptions in more extreme observations. The Q-Q plot on the right side of Fig. 5 is a graphical tool for assessing whether the residuals follow a normal distribution by comparing the quantiles of the sample residuals to the theoretical quantiles from a standard normal distribution. The points largely follow the straight 45-degree reference line in the center of the plot, which indicates that the central part of the residuals conforms well to the normal distribution. However, deviations from the line are visible at the

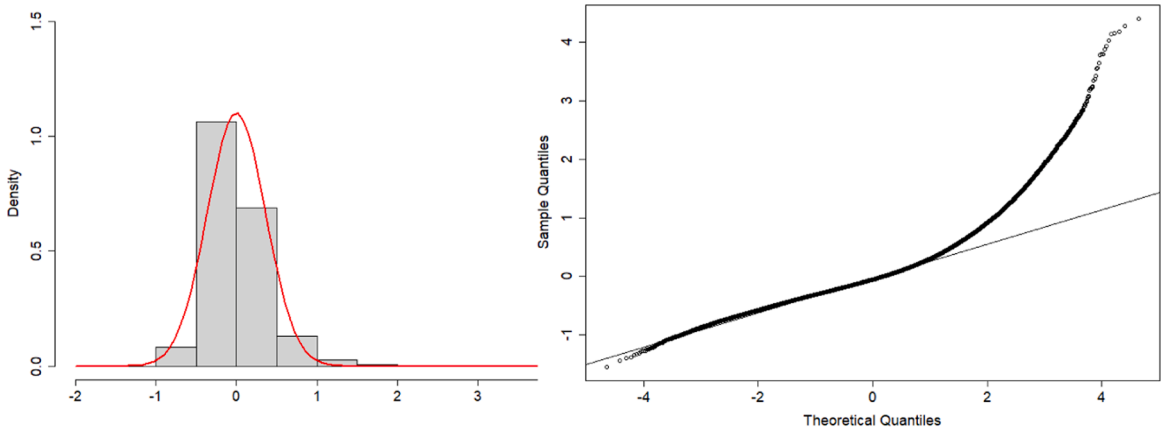

Fig. 5. Histogram of residuals and QQ-norm for Full model with impact and readership at time 0.

extremes of the distribution, in particular with upper quantiles showing upward curvature, suggesting that the residuals are not perfectly normally distributed. These departures, particularly in the extremes, are indicative of potential outliers or heavy-tailed behavior, which may affect the validity of inferences made using the model.

Therefore, we applied a bootstrap procedure. Bootstrapping is a non-parametric technique that allows for the estimation of the distribution of the model coefficients by resampling the observed data. In cases where errors deviate from normality, traditional coefficient estimates can become less reliable, and bootstrapping offers an effective approach to obtain more robust estimates and to assess the variability of the coefficients more accurately. For example, we applied this procedure to the full model with the "impact" variable at time t_0. Through the bootstrap method, for each model coefficient, we obtained two values:

- *Bias*, i.e. the difference between the average of the bootstrap estimates and the original estimate, indicating any potential deviation. Bias values near zero suggest that the bootstrap estimates align well with the original estimates.
- *Standard Error*, i.e. the standard deviation of the coefficients obtained from the bootstrap samples, providing an empirical measure of the variability of the estimates.

As evidenced in the supplemental material (SI-Bootstrap_results.xlsx), the values of *Bias* and *Standard Error* obtained for 500 iterations/samples are minimal and thus acceptable, confirming the stability of the estimates and the validity of the model.

Finally, in Fig. 6, we show the distribution of coefficients recorded for the 247 dummies related to as many SCs for each citation time window. The temporal pattern is evident, with frequency distributions tending to shift to higher values as the citation time window increases. For t_0, the coefficients are all invariably negative and the mean is -0.4, for t_1 the coefficient is positive for 6.5 % of all 247 SCs and the average increase to -0.175. For t_6, the coefficient is positive for 42.5 % of SCs, and the average becomes nihil. The variability of the distributions (evidenced by the shown interquartile ranges) tends to decrease with the citation time window.

4.4.2. Multicollinearity and model validity testing

In order to check for multicollinearity of variables and ensure the stability of the model, we computed Variance Inflation Factor (VIF) values using a version of the model that includes a constant term.[4] Table 6 reports diagnostic statistics for the Full models across all seven time points (from $t = 0$ to $t = 6$). The table includes VIF values for each explanatory variable, as well as Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) values.

As shown, all VIF values are well below the commonly accepted threshold of 10, indicating no problematic multicollinearity. Importantly, the values remain highly stable across time windows for most variables. This consistency reflects the fact that, aside from *L_IMPACT_t* and *L_READ_t*, all other predictors remain constant across the models (e.g., *L_AUTH, D_ENG, D_OPEN*, etc.). Therefore, the observed variation in VIF values over time is driven exclusively by the dynamic nature of the time-dependent variables *L_IMPACT_t* and *L_READ_t*, whose influence increases as the citation window expands.

Regarding model fit, both AIC and BIC decrease steadily from $t = 0$ to $t = 6$, suggesting a progressive improvement in model performance as early citation measures become stronger predictors. This trend supports the central argument of the paper: as the citation time window lengthens, the weight of non-citation factors diminishes, while early citations become increasingly dominant in

[4] Our models was theoretically developed without a constant, based on the rationale that when all independent variables are zero, the dependent variable must also be zero. In such cases, VIF values can become unreliable or misleading. In fact, omitting the constant term from a regression model can lead to biased or unstable VIF estimates, as the decomposition of variance becomes distorted in the absence of a proper baseline. Including the constant ensures that the VIF captures the true extent of shared variance among predictors in a realistic model setting. For this reason, the VIF values shown in Table 6 were calculated on the model including the constant, as this approach provides a more statistically robust and reliable measure of multicollinearity.

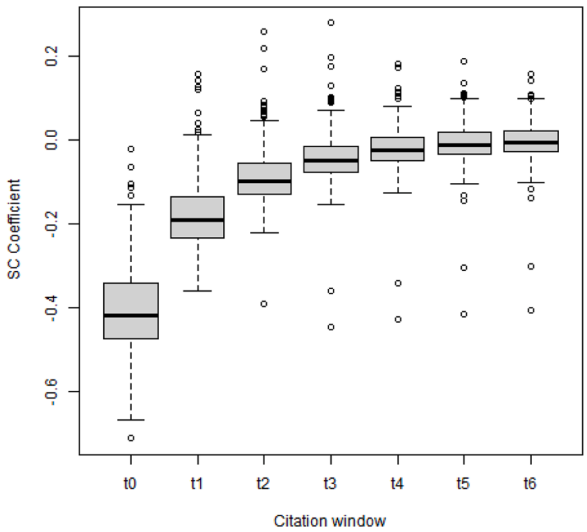

Fig. 6. Box plot of coefficients of SC dummies for the FULL model, by citation time window.

Table 6
– Diagnostic statistics for the Full models ($t = 0$ to $t = 6$), including VIF values for each variable, with the constant included, along with AIC and BIC.

Variables	$t = 0$	$t = 1$	$t = 2$	$t = 3$	$t = 4$	$t = 5$	$t = 6$
L_IMPACT_t	2.326968	1.803167	4.231894	5.010904	5.541096	5.928133	6.227836
L_READ_t	2.044624	2.044624	4.634625	5.467216	6.029057	6.434765	6.763226
L_AUTH	1.453024	1.453024	1.474544	1.474548	1.672232	1.480641	1.492335
D_ENG	1.670514	1.670514	1.672801	1.672570	1.672232	1.671948	1.671812
D_FOREIGN	1.715176	1.71517	1.715897	1.715928	1.715822	1.715694	1.252935
D_FUNDING	1.353621	1.353621	1.354842	1.354224	1.353614	1.352949	1.352407
D_OPEN	1.250005	1.250005	1.253853	1.253865	1.253541	1.253525	1.252935
L_PAGES	2.326968	2.326968	2.327975	2.329243	2.330709	2.331868	2.332545
L_IF	1.523991	1.523991	1.619340	1.621904	1.619263	1.612804	1.605816
D_ART	2.313529	2.313529	2.312878	2.314108	2.314786	2.315109	2.314771
D_REW	2.323133	2.323133	2.338233	2.341592	2.342975	2.344664	2.345353
L_REFER	2.450123	2.450123	2.519007	2.526159	2.528162	2.526610	2.524157
AIC*	237,677.2	104,987.9	−22,737.91	−132,856.8	−233,152.7	−330,334.7	−424,865.6
BIC	240,430.5	107,741.3	−19,984.57	−130,103.5	−230,399.3	−327,581.4	−422,112.2

* AIC and BIC refer to models estimated without the constant.

predicting long-term impact.

Table 7 presents diagnostic statistics for the Reduced models across the seven time points (from $t = 0$ to $t = 6$). As in the Full model, the table includes Variance Inflation Factor (VIF) values for each variable, calculated with the constant included, along with Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) values, derived from models estimated without the constant.

The VIF values in the Reduced model remain consistently low across all time points, with values well below 2 for both predictors (*L_IF_t* and *L_IMPACT_t*), indicating an absence of multicollinearity. These values are expectedly lower than those in the Full model, as the reduced specification includes only a limited number of explanatory variables, minimizing redundancy in the design matrix.

With respect to model fit, the AIC and BIC values decrease progressively as the citation time window increases, mirroring the trend observed in the Full model. The AIC improves substantially from 289,013.7 at $t = 0$ to –417,787.9 at $t = 6$, confirming the growing predictive power of early citations as time progresses. This further supports that early citation-based metrics (e.g., *L_IMPACT_t*) become

Table 7
– Diagnostic statistics for the Reduced models ($t = 0$ to $t = 6$), including VIF values for each variable, with the constant included, along with AIC and BIC.

Variables	$t = 0$	$t = 1$	$t = 2$	$t = 3$	$t = 4$	$t = 5$	$t = 6$
L_IF_t	1.212156	1.362402	1.412844	1.426465	1.429128	1.425852	1.420592
L_IMPACT_t	1.095770	1.238292	1.279779	1.289880	1.289880	1.287091	1.281796
AIC*	289,013.7	136,619.2	−3370.7	−11,9501.6	−222,525.2	−321,686.1	−417,787.9
BIC	291,661.2	139,266.6	−723.2	−11,6854.2	−219,877.8	−319,038.7	−415,140.4

* AIC and BIC refer to models estimated without the constant.

increasingly predictive of long-term impact over time, even in the absence of non-citation-based explanatory variables.

To assess the consistency of our results across publication years, we examined the descriptive statistics of the explanatory variables by year. The analysis revealed that the distributions of the key variables remain stable over time, with no notable differences in means, standard deviations, or ranges that would justify the inclusion of year-specific controls. Nevertheless, we re-estimated all 14 regressions (full and reduced models across the 7 citation windows), this time including year fixed effects. Although these effects are often statistically significant, their magnitude is negligible, with coefficients typically below the fourth or fifth decimal place. Furthermore, their signs are inconsistent: in the full model, year coefficients are negative in 5 out of 7 time windows, whereas in the reduced model they are consistently positive.

This combination of sign instability and minimal effect size suggests that including year fixed effects does not improve model interpretability or robustness and may instead introduce unnecessary noise or collinearity, particularly given the already time-structured nature of several explanatory variables. These findings support our methodological choice to base the analysis on the full three-year publication period, which we consider the most robust and balanced approach for evaluating long-term publication impact.

5. Discussion and conclusions

This study sought to enhance the prediction of long-term scholarly impact by integrating early citation data with Mendeley readership metrics and various non-scientific factors, such as journal impact factor, authorship and reference list characteristics, and publication accessibility. Our findings reveal that when combined with these additional attributes, early citation trends and readership data offer modest but statistically significant improvements in predicting a publication's long-term influence, thereby advancing models that traditionally rely on citations alone, with short time windows.

Our findings confirm that early citations, particularly within the first few years post-publication, remain among the strongest indicators of future impact. This result is consistent with earlier studies (Bornmann & Daniel, 2010), reinforcing the predictive value of early scholarly attention.

In addressing our first research question—what role do features beyond early citations play in predicting a publication's long-term impact—we found that integrating non-scientific factors offers modest but statistically significant improvements in predictive accuracy, especially in the early years post-publication. These factors, including Mendeley readership, journal impact factor, and authorship characteristics, provide complementary information beyond what early citations alone can offer, particularly in the 0–2 years after publication. The proposed model consistently outperforms prior models that include only early citations and IF (Abramo et al., 2019). This is especially evident in citation windows from 0 to 2 years, where the enhanced model achieves more precise predictions of long-term impact. These results suggest that while early citations are invaluable, they alone cannot fully capture the range of factors contributing to a publication's sustained influence. Among the non-scientific factors, readership metrics on Mendeley proved particularly valuable. Early readership correlates strongly with subsequent citations, suggesting that readership captures a publication's broader visibility and engagement beyond immediate scholarly circles. Journal impact factor and authorship characteristics, including international collaboration and native English-speaking authorship, also emerged as relevant predictors, although their explanatory power decreases as early citation data becomes more informative.

It should be noted that some models' outcomes seem to contradict previous literature. For instance, regression coefficients for the L_AUTH variable obtained for longer citation windows are negative, suggesting an effect of the number of authors on the publications' impact opposite with respect to most studies (Thelwall et al., 2023; Wu, Wang & Evans, 2019). In fact, this is due to the specificity of our model intentionally designed to introduce the lagged dependent variable (L_IMPACT_t) iteratively, thereby capturing the cumulative nature of citation dynamics. As expected in such models, the inclusion of increasingly informative lag terms absorbs explanatory power from certain predictors, leading to adjustments in the magnitude—and occasionally the sign—of their coefficients.

Concerning the second question—how the influence of these predictors changes with the length of the citation time window—our results show that the relative contribution of non-scientific factors appears to decrease over time, likely due to the increasing sufficiency of citation-based predictors. This finding underscores the time-sensitive nature of certain predictors, such as open access and funding, whose influence wanes as publications age. The pattern highlighted—that non-citation variables offer greater predictive value in the early years, while their incremental contribution becomes less substantial after year three—is indeed consistent with both theoretical expectations and empirical patterns observed in the literature.

In the early stages of a publication's life, when citation counts are still accumulating, and thus volatile or sparse, non-citation factors (e.g., journal prestige, Mendeley readership, collaboration patterns, and reference list length) play a crucial role in approximating a publication's potential impact. These variables serve as proxies for visibility, credibility, and scholarly engagement, which are particularly informative when citation-based signals are still weak.

However, as time progresses, early citation counts themselves become increasingly reliable indicators of long-term impact. Citations begin to stabilize and reflect the actual scholarly reception of the work. At this point (i.e., after year three), the predictive value of non-citation variables becomes relatively marginal because much of the variance in future citations is already captured by the accumulated early citations.

Fig. 3 clearly illustrates this shift in explanatory power. The dominance of early citation indicators after three years is not a sign of the irrelevance of non-citation factors, but rather a reflection of the evolving information landscape: early citations eventually absorb much of the explanatory contribution that non-citation variables initially provided.

Finally, regarding the third research question—how much the predictive power of early citations improves when these additional predictors are considered—we demonstrated that the full model, which integrates all factors, consistently outperforms reduced models

relying solely on early citations or on early citations and journal impact factor. The improvement is most significant for shorter citation windows, suggesting that incorporating broader features is particularly valuable when long-term citation data are not yet available. Theoretically, this study contributes to the understanding of scientific impact prediction by validating that integrating non-citation-based factors with early citation counts offers a more comprehensive, though still citation-centered, prediction of a publication's long-term influence. Traditional citation-based models focus on scholarly attention alone, potentially overlooking non-scholarly engagement and visibility metrics that offer additional insight into a publication's reach and significance. By including non-scientific attributes, we offer a model that accounts for a broader range of influence factors, recognizing the contribution of both scholarly responses and external factors to long-term influence.

Practically, these findings have potential implications for stakeholders such as research funders, institutions, and individual researchers—particularly in research systems similar to the Italian context from which our data are drawn. While we believe the insights may be informative and supportive of more informed decisions about funding allocations and research priorities, they should be interpreted in light of the specific institutional and disciplinary characteristics of the dataset. For funding bodies and policymakers operating in comparable environments and scholars working in contexts with similar research dynamics, the enhanced model may offer a significant refinement in the predictive value of early indicators of publication impact.

In addition, this study's findings open potential pathways for enhancing research evaluation practices in institutional and funding agency settings. By adopting a more comprehensive model that includes both citation and non-citation factors, evaluators—particularly those operating in systems resembling the Italian academic landscape—may be able to assess emerging research with greater precision, capturing the broader spectrum of factors that predict influence beyond citations alone.

Most existing bibliometric studies assess the impact of publications solely by counting citations, without considering the citation time window. The novelty of our work lies in its comprehensive integration of early citations and publication characteristics to derive readily usable predictor coefficients. These coefficients can assist researchers and practitioners in more accurately assessing the impact of publications. They are reported per year and subject category in the supplementary materials.

While our study advances impact prediction methods, it is important to acknowledge certain limitations. First, our analysis focused on publications from Italy within a specific timeframe (2010–2012), which may limit the generalizability of the findings to other contexts. It is plausible that the importance of specific predictors varies across fields and countries due to differences in publication practices, collaboration networks, and access to funding. Future research could extend this work by testing the model on data from diverse regions, publication years, and scientific disciplines to assess the robustness of these predictors.

Second, although Mendeley readership and citation metrics are strong indicators of academic engagement, their predictive power could potentially be improved by incorporating broader altmetrics. Social media platforms, news mentions, and policy document citations may capture additional layers of public and interdisciplinary interest not reflected in traditional citation data alone. As digital media increasingly shapes the dissemination of knowledge, future studies could examine how these alternative metrics contribute to long-term impact, especially in fields that engage heavily with non-academic audiences.

Another limitation involves the focus on quantitative predictors alone. Non-numeric characteristics, such as the novelty of research ideas, the clarity of communication, or the interdisciplinary nature of the work, are likely influential but challenging to quantify. Machine learning approaches, such as natural language processing (NLP) techniques, could be explored to assess text features like abstract complexity, thematic novelty, or interdisciplinarity. By integrating these content-driven insights, future models might offer an even more nuanced prediction of long-term influence.

In summary, our study demonstrates that combining early citation metrics with a range of non-citation-based factors results in improved predictions of a publication's long-term impact. This approach not only expands traditional models but also offers a more nuanced understanding of how publications achieve influence over time. By capturing both scholarly and non-scholarly dimensions of impact, our model provides a practical, accessible framework for assessing research influence in its early stages, empowering funders, institutions, and researchers to make better-informed decisions about the potential value of emerging research.

As the scholarly landscape continues to evolve, this integrated model represents a promising step toward more timely, comprehensive research evaluation tools that align with the multifaceted nature of scientific impact.

CRediT authorship contribution statement

Giovanni Abramo: Writing – review & editing, Writing – original draft, Supervision, Investigation, Conceptualization. **Tindaro Cicero:** Writing – original draft, Visualization, Methodology, Investigation, Data curation. **Ciriaco Andrea D'Angelo:** Writing – original draft, Visualization, Methodology, Investigation, Data curation, Conceptualization.

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Supplementary materials

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References

Abramo, G., & D'Angelo, C. A (2015). The relationship between the number of authors of a publication, its citations and the impact factor of the publishing journal: Evidence from Italy. *Journal of Informetrics, 9*(4), 746–761.

Abramo, G., D'Angelo, C. A., & Di Costa, F. (2016). The effect of a country's name in the title of a publication on its visibility and citability. *Scientometrics, 109*(3), 1895–1909.

Abramo, G., D'Angelo, C. A., & Di Costa, F. (2017). Do interdisciplinary research teams deliver higher gains to science? *Scientometrics, 111*(1), 317–336.

Abramo, G., D'Angelo, C. A., & Felici, G. (2019). Predicting long-term publication impact through a combination of early citations and journal impact factor. *Journal of Informetrics, 13*(1), 32–49.

Abramo, G. (2018). Revisiting the scientometric conceptualization of impact and its measurement. *Journal of Informetrics, 12*(3), 590–597.

Acuna, D. E., Allesina, S., & Kording, K. P. (2012). Future impact: Predicting scientific success. *Nature, 489*(7415), 201–202.

Aksnes, D. W. (2003). Characteristics of highly cited papers. *Research Evaluation, 12*(3), 159–170.

Aksnes, D. W. (2006). Citation rates and perceptions of scientific contribution. *Journal of the American Society for Information Science and Technology, 57*(2), 169–185.

Alimohammadi, D., & Sajjadi, M. (2009). Correlation between references and citations. *Webology, 6*(2), a71.

Ammon, U. (2012). Linguistic inequality and its effects on participation in scientific discourse and exchange of knowledge. *Language Problems and Language Planning, 36*(3), 193–214.

Bai, X., Liu, H., Zhang, F., Ning, Z., Kong, X., Lee, I., & Xia, F. (2017). An overview on evaluating and predicting scholarly article impact. *Information, 8*(3), 73.

Ball, P. (2008). A longer paper gathers more citations. *Nature, 455*(7211), 274.

Bornmann, L., & Daniel, H. D. (2008). What do citation counts measure? A review of studies on citing behavior. *Journal of Documentation, 64*(1), 45–80.

Bornmann, L., & Daniel, H. D. (2010). The citation speed index: A useful bibliometric indicator to add to the h index. *Journal of Informetrics, 4*(3), 444–446.

Bornmann, L., & Leydesdorff, L. (2017). Skewness of citation impact data and covariates of citation distributions: A large-scale empirical analysis based on Web of Science data. *Journal of Informetrics, 11*(1), 164–175.

Boyack, K. W., & Jordan, P. (2011). Comparing alternative institutional rankings based on citation data. *Journal of the American Society for Information Science and Technology, 62*(11), 2181–2194.

Costas, R., Perianes-Rodríguez, A., & Ruiz-Castillo, J. (2017). On the quest for currencies of science: Field "exchange rates" for citations and Mendeley readership. *Aslib Journal of Information Management, 69*(5), 557–575.

De Santis Puzzonia, M., Beltram, F., Cicero, T., & Malgarini, M. (2018). Accounting for co-authorship in research evaluation: The case of experimental physics in Italy. In *STI 2018 Conference Proceedings* (pp. 967–974). Centre for Science and Technology Studies (CWTS).

de Solla Price, D. (1976). A general theory of bibliometric and other cumulative advantage processes. *Journal of the Association for Information Science and Technology, 27*, 292–306.

Eom, Y.-H., & Fortunato, S. (2011). Characterizing and modeling citation dynamics. *PLoS ONE, 6*(9), Article e24926.

Fang, Z., Ho, C., Han, Z., & Wu, P. (2024). How fast do scholarly papers get read by various user groups? A longitudinal and cross-disciplinary analysis of the evolution of Mendeley readership. *Journal of the Association for Information Science and Technology, 75*(13), 1433–1450.

Foster, J. G., Rzhetsky, A., & Evans, J. A. (2015). Tradition and innovation in scientists' research strategies. *American Sociological Review, 80*(5), 875–908.

Fox, C. W., Paine, C. T., & Sauterey, B. (2016). Citations increase with manuscript length, author number, and references cited in ecology journals. *Ecology and Evolution, 6*(21), 7717–7726.

Garfield, E. (2006). The history and meaning of the journal impact factor. *JAMA, 295*(1), 90–93.

Gargouri, Y., Hajjem, C., Larivière, V., Gingras, Y., Carr, L., Brody, T., et al. (2010). Self-selected or mandated, open access increases citation impact for higher-quality research. *PLoS ONE, 5*(10).

Glänzel, W., & De Lange, C. (2002). A distributional approach to multinationality measures of international scientific collaboration. *Scientometrics, 54*, 75–89.

Haustein, S., Larivière, V., Thelwall, M., Amyot, D., & Peters, I. (2014). Tweets vs. Mendeley readers: How do these two social media metrics differ? *IT - Information Technology, 56*(5), 207–215.

Heßler, N., & Ziegler, A. (2022). Evidence-based recommendations for increasing the citation frequency of original articles. *Scientometrics, 127*, 3367–3381.

Himani, S., Kumar, M. H., Enduri, M. K., Begum, S. S., Rageswari, G., & Anamalamudi, S. (2022). A comparative study on machine learning based prediction of citations of articles. Paper presented at the. In *2022 6th International Conference on Trends in Electronics and Informatics, ICOEI 2022 - Proceedings* (pp. 1819–1824).

Hou, J., Pan, H., Guo, T., Lee, I., Kong, X., & Xia, F. (2019). Prediction methods and applications in the science of science: A survey. *Computer Science Review, 34*, Article 100197.

Hyland, K. (2016). Academic publishing and the myth of linguistic injustice. *Journal of Second Language Writing, 33*, 43–54.

Jiang, J., He, D., & Ni, C. (2013). The correlations between article citation and references' impact measures: What can we learn? *Proceedings of the American society for information science and technology, 50*(1), 1–4.

Kosteas, V. D. (2020). Predicting the future impact of scientific papers based on early citations. *Journal of Informetrics, 14*(4), Article 101068.

Kousha, K., & Thelwall, M. (2024). Factors associating with or predicting more cited or higher quality journal articles: An Annual Review of Information Science and Technology (ARIST) paper. *Journal of the Association for Information Science and Technology, 75*(3), 15–44.

Larivière, V., & Gingras, Y. (2010). On the relationship between interdisciplinarity and scientific impact. *Journal of the Association for Information Science & Technology, 61*(1), 126–131.

Leahey, E., Lee, J., & Funk, R. J. (2023). What types of novelty are most disruptive? *American Sociological Review, 88*(3), 562–597.

Levitt, J. M., & Thelwall, M. (2011). A combined bibliometric indicator to predict article impact. *Information Processing and Management, 47*(2), 300–308.

Lewison, G., & Dawson, G. (1998). The effect of funding on the outputs of biomedical research. *Scientometrics, 41*(1–2), 17–27.

Li, G., Xie, Y., Wang, Y., & Xie, J. (2019). Prediction of citation counts based on deep neural networks. *Journal of Informetrics, 13*(1), 407–418.

Liu, J., Chen, H., Liu, Z., Bu, Y., & Gu, W. (2022). Non-linearity between referencing behavior and citation impact: A large-scale, discipline-level analysis. *Journal of Informetrics, 16*(3), Article 101318.

Mammola, S., Fontaneto, D., Martínez, A., & Chichorro, F. (2021). Impact of the reference list features on the number of citations. *Scientometrics, 126*(1), 785–799.

Mammola, S., Piano, E., Doretto, A., Caprio, E., & Chamberlain, D. (2022). Measuring the influence of non-scientific features on citations. *Scientometrics, 127*(7), 4123–4137.

Mazloumian, A., Eom, Y.-H., Helbing, D., Lozano, S., & Fortunato, S. (2011). How citation boosts promote scientific paradigm shifts and Nobel Prizes. *PLoS ONE, 6*(5), Article e18975.

Meneghini, R., & Packer, A. L. (2007). Is there science beyond English? Initiatives to increase the quality and visibility of non-English publications might help to break down language barriers in scientific communication. *EMBO Reports, 8*(2), 112–116.

Moed, H. F. (2005). *Citation analysis in research evaluation*. Springer Science & Business Media.

Mohammadi, E., & Thelwall, M. (2014). Mendeley readership altmetrics for the social sciences and humanities: Research evaluation and knowledge flows. *Journal of the Association for Information Science and Technology, 65*(8), 1627–1638.

Narin, F., & Whitlow, E. S. (1990). Measurement of Scientific Co-operation and Coauthorship in CEC-related areas of Science. *Report EUR 12900*. Luxembourg: Office for Official Publications in the European Communities.

Newman, M. (2018). *Models of network formation* (pp. 434–492). Oxford Academic. Networks, 2nd edn, Oxford, 2018; online edn.

Nicolaisen, J., & Frandsen, T. F. (2021). Number of references: A large-scale study of interval ratios. *Scientometrics, 126*(1), 259–285.

Özkent, Y. (2022). Social media usage to share information in communication journals: An analysis of social media activity and article citations. *PLoS ONE, 17*(2), Article e0263725.

Parolo, P. D. B., Pan, R. K., Ghosh, R., Huberman, B. A., Kaski, K., & Fortunato, S. (2015). Attention decay in science. *Journal of Informetrics, 9*(4), 734–745.

Penner, O., Pan, R. K., Petersen, A. M., Kaski, K., & Fortunato, S. (2013). On the predictability of future impact in science. *Scientific Reports, 3*, 3052.

Petersen, A. M., Fortunato, S., Pan, R. K., Kaski, K., Penner, O., & Pammolli, F. (2014). Reputation and impact in academic careers. *Proceedings of the National Academy of Sciences, 111*(43), 15316–15321.
Porter, A. L., & Rafols, I. (2009). Is science becoming more interdisciplinary? Measuring and mapping six research fields over time. *Scientometrics, 81*, 719–745.
Qiu, J., & Han, X. (2024). An early evaluation of the long-term influence of academic papers based on machine learning algorithms. *IEEE access : Practical innovations, open solutions, 12*, 41773–41786.
Rigby, J. (2013). Looking for the impact of peer review: Does funding help to explain higher citation rates? *Scientometrics, 94*(1), 57–73.
Rossi, M. J., & Brand, J. C. (2020). Journal article titles impact their citation rates. *Arthroscopy : The journal of arthroscopic & related surgery : Official publication of the Arthroscopy Association of North America and the International Arthroscopy Association, 36*, 2025–2029.
Ruan, X., Zhu, Y., Li, J., & Cheng, Y. (2020). Predicting the citation counts of individual papers via a BP neural network. *Journal of Informetrics, 14*(3), Article 101039.
Sanfilippo, P., Hewitt, A. W., & Mackey, D. A. (2018). Plurality in multi-disciplinary research: Multiple institutional affiliations are associated with increased citations. *PeerJ*, (6), e5664.
Sarigöl, E., Pfitzner, R., Scholtes, I., Garas, A., & Schweitzer, F. (2014). Predicting scientific success based on coauthorship networks. *EPJ Data Science, 3*(9).
Sivadas, E., & Johnson, M. S. (2015). Relationships between article references and subsequent citations of marketing journal articles. *Revolution in marketing: Market driving changes* (pp. 199–205). Cham: Springer.
Tahamtan, I., Safipour Afshar, A., & Ahamdzadeh, K. (2016). Factors affecting number of citations: A comprehensive review of the literature. *Scientometrics, 107*(3), 1195–1225.
Talaat, F. M., & Gamel, S. A. (2023). Predicting the impact of no. of authors on no. of citations of research publications based on neural networks. *Journal of Ambient Intelligence and Humanized Computing, 14*, 8499–8508.
Thelwall, M., Kousha, K., Abdoli, M., Stuart, E., Makita, M., Wilson, P., & Levitt, J. (2023). Why are coauthored academic articles more cited: Higher quality or larger audience? *Journal of the Association for Information Science and Technology, 74*(7), 791–810.
Traag, V. A. (2021). Inferring the causal effect of journals on citations. *Quantitative Science Studies, 2*(2), 496–504.
Uzzi, B., Mukherjee, S., Stringer, M., & Jones, B. (2013). Atypical combinations and scientific impact. *Science (New York, N.Y.), 342*(6157), 468–472.
van Leeuwen, T., & Moed, H. F. (2002). Language bias in the coverage of the Science Citation Index and its consequences for international comparisons of national research performance. *Scientometrics, 51*(1), 335–346.
Wang, D., Song, C., & Barabási, A. (2013). Quantifying long-term scientific impact. *Science (New York, N.Y.), 342*(6154), 127–132.
Wang, X., Liu, C., Mao, W., & Fang, Z. (2015). The open access advantage considering citation, article usage and social media attention. *Scientometrics, 103*(2), 555–564.
Wang, J., Veugelers, R., & Stephan, P. (2017). Bias against novelty in science: A cautionary tale for users of bibliometric indicators. *Research Policy, 46*(8), 1416–1436.
Wang, K., Shi, W., Bai, J., et al. (2021). Prediction and application of article potential citations based on nonlinear citation-forecasting combined model. *Scientometrics, 126*(8), 6533–6550.
Wang, J., Thijs, B., & Glänzel, W. (2015). Interdisciplinarity and impact: Distinct effects of variety, balance, and disparity. *PLoS ONE, 10*(5).
Wu, L. F., Wang, D. S., & Evans, J. A. (2019). Large teams develop and small teams disrupt science and technology. *Nature, 566*(7744), 378–382.
Wuchty, S., Jones, B. F., & Uzzi, B. (2007). The increasing dominance of teams in production of knowledge. *Science (New York, N.Y.), 316*(5827), 1036–1039.
Xia, W., Li, T., & Li, C. (2023). A review of scientific impact prediction: Tasks, features and methods. *Scientometrics, 128*(1), 543–585.
Xie, J., Gong, K., Li, J., Ke, Q., Kang, H., & Cheng, Y. (2019). A probe into 66 factors which are possibly associated with the number of citations an article received. *Scientometrics, 119*(3), 1429–1454.
Yau, C. K., Porter, A., Newman, N., & Suominen, A. (2014). Clustering scientific documents with topic modeling. *Scientometrics, 100*, 767–786.
Yu, M., Wang, X., & Guo, H. (2014). Citation prediction using time series segmentation and machine learning. *Scientometrics, 101*(3), 1553–1569.
Yu, X., Meng, Z., Qin, D., Shen, C., & Hua, F. (2022). The long-term influence of open access on the scientific and social impact of dental journal articles: An updated analysis. *Journal of Dentistry, 119*, Article 104067.
Zahedi, Z., Costas, R., & Wouters, P. (2014). How well developed are altmetrics? A cross-disciplinary analysis of the presence of 'alternative metrics' in scientific publications. *Scientometrics, 101*(2), 1491–1513.
Zhang, F., & Wu, S. (2024). Predicting citation impact of academic papers across research areas using multiple models and early citations. *Scientometrics, 129*, 4137–4166.